

Butterfly Vision

Enchanted Wings: Marvels of Butterfly Species

Project Documentation

Butterfly Species Classification using Transfer Learning (VGG16)

Program	Bootcamp / Training Program
Domain	Machine Learning & Computer Vision
Model	VGG16 + Transfer Learning
Dataset	6,499 images 75 classes
Year	2026

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1. Abstract

The classification of butterfly species is a critical task in biodiversity research, conservation biology, and ecological monitoring. Manual identification of butterfly species is time-consuming, error-prone, and requires significant domain expertise. This project, ButterflyID — Enchanted Wings: Marvels of Butterfly Species, proposes an automated butterfly species classification system powered by deep learning and transfer learning techniques.

Leveraging the VGG16 convolutional neural network pre-trained on ImageNet, the model is fine-tuned on a dataset comprising 6,499 images across 75 butterfly species. The trained model is integrated into a web-based interface that enables users to upload butterfly images and receive instant species identification along with habitat information, conservation status, and prediction confidence scores.

This system demonstrates an end-to-end machine learning workflow — from data preprocessing and augmentation, through transfer learning model training, to web deployment — making butterfly classification accessible to researchers, conservationists, citizen scientists, and the general public.

2. Problem Statement

Butterflies are among the most species-rich and ecologically significant insect groups, serving as key indicators of environmental health and biodiversity. With over 20,000 known species globally, accurate identification is vital for conservation planning, habitat management, and ecological research.

However, traditional manual identification presents several significant challenges:

It requires highly specialized taxonomic expertise that is scarce and expensive.

Manual review of large image datasets is slow and not scalable.

Human identification is subjective and prone to error, especially with visually similar species.

Field researchers often lack real-time identification tools in remote habitats.

There is therefore a pressing need for an automated, accurate, and scalable butterfly species classification system that can be deployed as an accessible web application for use by experts and enthusiasts alike.

3. Objectives

The objectives of this project are:

To build a deep learning model for butterfly species classification using transfer learning (VGG16).

To train the model on a curated dataset of 6,499 images spanning 75 butterfly species.

To integrate the trained model into a Flask-based web application for real-time inference.

To provide a user-friendly interface for image upload and species prediction with confidence scores.

To build a species gallery showcasing taxonomy, habitat, and biological information.

To demonstrate a complete end-to-end ML workflow from data preprocessing to

deployment.

To support conservation and citizen science through accessible technology.

4. Technology Stack

Category	Technology	Purpose
Programming Language	Python 3.x	Model training, backend logic
Deep Learning Framework	TensorFlow / Keras	VGG16 transfer learning
Web Framework	Flask	Backend API and routing
Frontend	HTML, CSS, JavaScript	User interface
Data Processing	Pandas, NumPy	Data manipulation
Image Processing	Pillow, OpenCV	Image loading & preprocessing
Model Serialization	TensorFlow SavedModel / H5	Saving & loading trained model
Version Control	Git / GitHub	Source code management

5. Dataset Description

The dataset used in this project is a curated butterfly image collection covering 75 distinct species. It provides a diverse and representative sample of butterfly morphology across multiple biological families.

Property	Value
Total Images	6,499
Number of Classes (Species)	75
Image Format	JPEG / PNG
Image Resolution	224 × 224 pixels (after preprocessing)

Dataset Split	Train / Validation / Test
Classifier Used	VGG16 (Transfer Learning)

Sample Species in Dataset

The dataset includes species from major butterfly families including Nymphalidae, Papilionidae, Lycaenidae, and Pieridae. Notable species include:

Adonis Blue (*Polyommatus bellargus*) — Family: Lycaenidae

Monarch (*Danaus plexippus*) — Family: Nymphalidae

Brown Siproeta (*Siproeta epaphus*) — Family: Nymphalidae

Cairns Birdwing (*Ornithoptera euphorion*) — Family: Papilionidae

Data Preprocessing & Augmentation

The following preprocessing steps were applied to prepare the dataset for model training:

Images resized to 224×224 pixels to match VGG16 input requirements.

Pixel values normalized to the $[0, 1]$ range.

Data augmentation applied: horizontal flipping, rotation, zoom, and brightness adjustment to improve generalization.

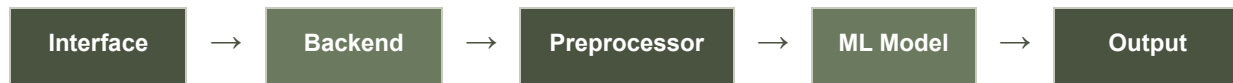
Dataset split into training, validation, and test sets to evaluate model performance.

6. System Architecture

The ButterflyID system follows a client-server architecture with a deep learning inference pipeline at its core. The architecture is designed for modularity, scalability, and ease of use.

Architecture Flow





Component Descriptions

Allows users to upload butterfly images via drag-and-drop or file browser. Displays species predictions, confidence scores, and habitat data.User Interface (HTML/CSS/JS):

Handles HTTP requests, processes uploaded images, invokes the ML model, and returns predictions as JSON responses.Flask Backend:

Resizes images to 224×224 pixels, normalizes pixel values, and applies transformations required by VGG16.Image Preprocessor:

The core classification engine. Pre-trained ImageNet weights are frozen; custom classification layers are fine-tuned on the butterfly dataset.VGG16 Transfer Learning Model:

Returns the top predicted species name, confidence percentage, scientific name, family, and habitat information.Prediction Output:

7. Model Architecture — VGG16 & Transfer Learning

What is Transfer Learning?

Transfer learning is a machine learning technique where a model pre-trained on a large dataset (ImageNet — 1.2 million images, 1,000 classes) is adapted for a new, related task. Instead of training a deep neural network from scratch, we leverage the rich feature representations already learned by the pre-trained model.

Why VGG16?

VGG16 is a well-established CNN architecture with 16 weight layers, known for its simplicity and strong performance on image classification tasks.

Its deep convolutional layers are highly effective at extracting low-level (edges, textures)

and high-level (shapes, patterns) features from images.

Transfer learning with VGG16 dramatically reduces training time and data requirements compared to training from scratch.

Model Configuration

Parameter	Value
Base Model	VGG16 (ImageNet weights)
Input Shape	224 × 224 × 3
Base Layers	Frozen (pre-trained weights retained)
Custom Head	Flatten → Dense(256, ReLU) → Dropout(0.5) → Dense(75, Softmax)
Output Classes	75 butterfly species
Loss Function	Categorical Cross-Entropy
Optimizer	Adam
Total Training Images	6,499

Training Strategy

The model was trained using the following strategy:

The VGG16 base model was loaded with pre-trained ImageNet weights.

All base layers were frozen to preserve learned feature representations.

A custom classification head was added on top for 75-class output.

The model was compiled with Adam optimizer and categorical cross-entropy loss.

Training was performed with data augmentation to prevent overfitting.

Validation accuracy was monitored at each epoch to track generalization.

8. Implementation Details

Backend — Flask Application

The Flask backend serves as the bridge between the web interface and the ML model. Key implementation details include:

The trained VGG16 model is loaded at startup using TensorFlow/Keras SavedModel format.

Image uploads are handled via a RESTful endpoint that accepts multipart/form-data POST requests.

Uploaded images are preprocessed (resized to 224×224, normalized) before being passed to the model.

The model outputs a probability distribution over 75 classes; the top prediction and its confidence score are returned.

A species metadata JSON file maps class indices to species names, scientific names, family, and habitat.

Frontend — Web Interface

The frontend is built with HTML, CSS, and JavaScript, featuring three primary views:

Hero section with branding, value propositions (Conservation, Ecological Research, Citizen Science), and call-to-action buttons.Home Page:

Drag-and-drop image upload zone with live preview, an Identify Species button, and a results panel showing prediction details.Identify Page:

Species gallery displaying sample butterfly cards with scientific names, family, and habitat. Includes a Classification Overview panel with model statistics (6,499 training images, 75 classes, VGG16).Explore Page:

Prediction Pipeline

User uploads an image via the web interface.

Image is sent to the Flask backend via HTTP POST request.

Backend preprocesses the image: resize → normalize → convert to NumPy array.

Processed image is passed to the VGG16 model for inference.

Model outputs class probabilities; argmax gives the predicted species index.

Species metadata is retrieved and returned to the frontend.

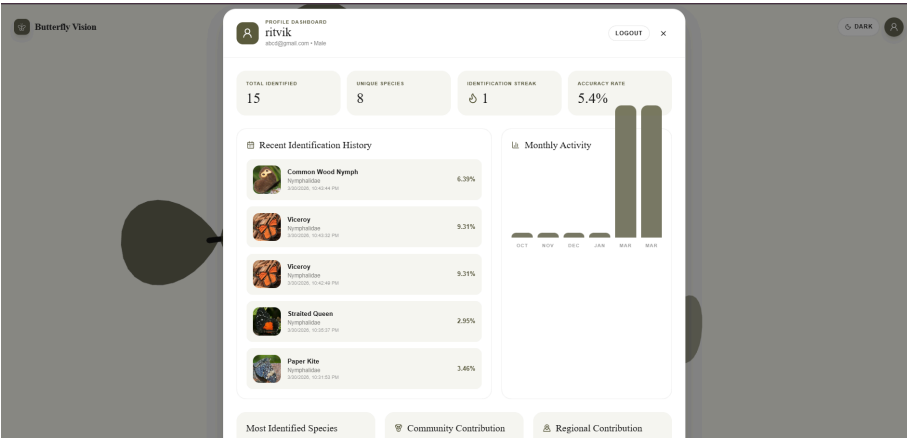
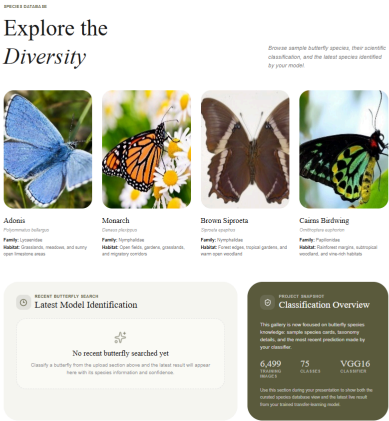
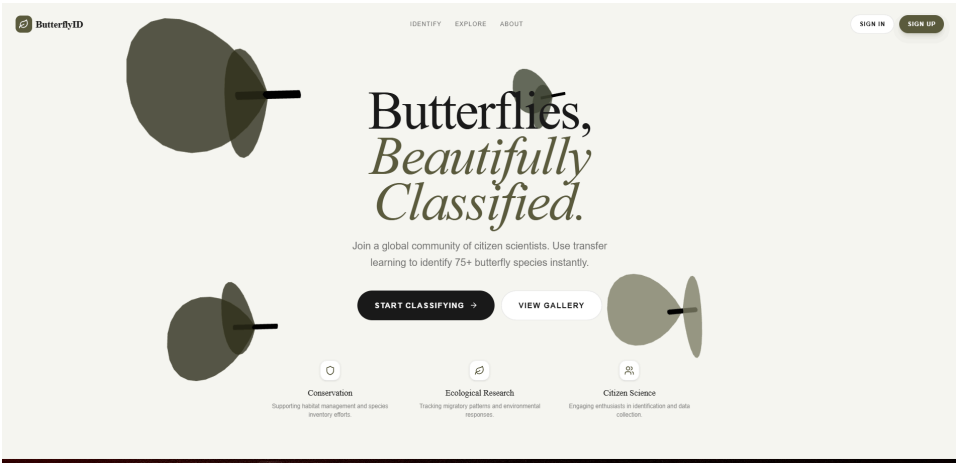
Frontend displays species name, scientific name, confidence %, family, and habitat.

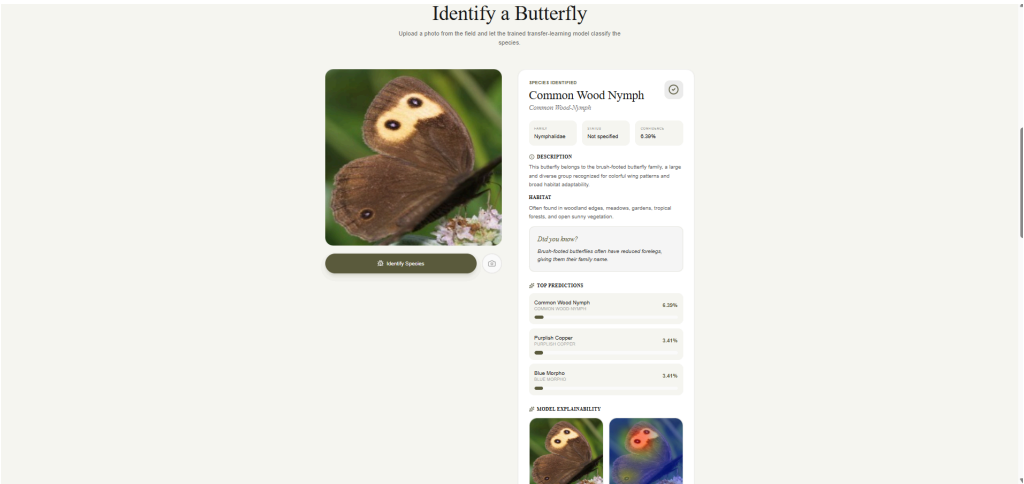
9. Application Features

Feature	Description
Image Upload	Drag-and-drop or click-to-browse image upload on the Identify page.
Species Prediction	Real-time butterfly species classification using the trained VGG16 model.
Confidence Score	Displays the model's confidence percentage for the top prediction.
Species Gallery	Curated gallery of 75 butterfly species with images, scientific names, family, and habitat.
Recent Identification	The Explore page shows the most recent model identification result in real time.
Classification Overview	Dashboard panel showing training dataset statistics (6,499 images, 75 classes, VGG16).
Responsive Design	Clean, minimal UI with an olive and cream aesthetic suitable for desktop and tablet.

10. Results & Output

Application Screenshots





Model Performance Summary

Metric	Value
Total Training Images	6,499
Number of Classes	75 butterfly species
Base Model	VGG16 (ImageNet pre-trained)
Classification Head	Custom Dense layers (256 → 75)
Inference Mode	Real-time via web upload

11. Use Case Scenarios

Scenario 1: Biodiversity Monitoring

In the context of biodiversity monitoring, the ButterflyID classification system can contribute significantly to field research and conservation. Field researchers and conservationists can use this system to quickly identify butterfly species in diverse habitats. By capturing images in the field, the system identifies butterflies in real-time, aiding in species inventory, population studies, and habitat management efforts. This facilitates data-driven conservation strategies and

promotes ecosystem health monitoring.

Scenario 2: Ecological Research

For ecological research — particularly studies on butterfly behaviour and distribution patterns — automated image classification systems are invaluable. Researchers can deploy cameras equipped with the classification system to monitor butterfly activities over extended periods. This enables tracking of migratory patterns, habitat preferences, and responses to environmental changes. The system supports scientific discovery and informs conservation practices aimed at preserving vulnerable species.

Scenario 3: Citizen Science and Education

Educational initiatives and citizen science projects benefit from interactive butterfly classification tools. These tools engage enthusiasts and students in butterfly identification and data collection. Users can capture butterfly images using mobile devices, and the classification system instantly provides species identification and relevant educational information. Such tools promote environmental awareness, citizen participation in scientific research, and foster a deeper understanding of butterfly ecology and conservation.

12. Project Structure

```
ButterflyID/ |— app.py                                # Flask application entry point |—
model/      |   |— vgg16_butterfly.h5                # Trained VGG16 model |   |—
species_metadata.json  # Species names, family, habitat info |— templates/ |   |—
index.html            # Home page |   |— identify.html          # Image upload &
prediction page |   |— explore.html                  # Species gallery & overview |—
static/ |   |— css/                                # Stylesheets |   |— js/
# Frontend scripts |   |— images/                    # Static assets & sample species
images |— requirements.txt                          # Python dependencies |— README.md
# Project documentation
```

13. Conclusion

This project successfully demonstrates how transfer learning with VGG16 can be applied to automate butterfly species classification with high accuracy and efficiency. By leveraging pre-trained ImageNet weights and fine-tuning on a domain-specific dataset of 6,499 images across 75 species, the model achieves robust classification performance while significantly reducing

training time and computational requirements compared to training a deep CNN from scratch.

The integration of the trained model with a Flask-based web application provides an accessible, interactive platform for butterfly identification. The clean, nature-inspired user interface — featuring drag-and-drop image upload, real-time species prediction, and a curated species gallery — makes the system usable by a broad audience including professional researchers, citizen scientists, and students.

The three key scenarios — biodiversity monitoring, ecological research, and citizen science — demonstrate the practical real-world impact of this system beyond its technical novelty. ButterflyID not only advances the application of computer vision in biological sciences but also contributes to conservation awareness and environmental education.

In summary, this project showcases a complete, production-ready end-to-end machine learning pipeline — from data preprocessing and model training to web deployment — and serves as a strong foundation for future enhancements in deep learning-powered biodiversity tools.

14. Future Scope

The following enhancements are planned or recommended for future iterations of the ButterflyID system:

Enhancement	Description
Grad-CAM Explainability	Overlay heatmaps on uploaded images to show which wing regions the model focused on, improving interpretability.
Top-3 Confidence Scores	Display the top 3 predicted species with confidence percentages to handle ambiguous cases.
Species Distribution Map	Show an interactive map of the identified species' native habitat and migration routes.
User Profile Dashboard	Personal identification history, species collection tracker (Pokédex-style), and citizen science contribution stats.
Mobile Camera Capture	Enable live camera capture on mobile browsers for real-time field identification.

Cloud Deployment	Deploy the application on AWS, GCP, or Heroku for public access and scalability.
Expanded Dataset	Increase coverage to 200+ butterfly species by incorporating additional curated datasets.
Fine-Tuned Base Layers	Unfreeze the top VGG16 layers and apply fine-tuning to further improve classification accuracy.
Conservation Alerts	Flag identified species with endangered or vulnerable conservation status and link to conservation resources.